HW1

```
aa <- read.delim("SRBCT_train.txt") # 讀檔
bb <- as.matrix(aa[,-c(1,2)]) # 扣掉第 1、2 col，因為是
character
dim(bb) # 觀察維度
```

Ouput:

```
[1] 2308   63
```

題目一：Get mean & sd value for each gene cross samples

Get mean value for each gene cross samples

```
# 求 means
# 方法一 -> 使用 apply
mean_gene_cross_samples1 <- apply(bb, 1, mean)
mean_gene_cross_samples1

# 方法二 -> 使用內建函式 rowMeans
mean_gene_cross_samples2 <- rowMeans(bb)
```

Ouput:

```
  [1] 1.4999175 0.2859635 0.9738683 0.6240492 0.3011079 2.1054048 5.0772444 1.2308952 1.2954397 1.5800095 5.4774159 0.8577889 5.0337540
 [14] 1.7690492 1.0853603 0.7313683 1.0727603 2.3230905 1.0906667 1.2494476 1.6026540 5.8988127 0.5117032 4.1544032 3.2013159 5.0632984
 [27] 1.1200556 1.3612857 0.3845222 1.2235825 0.5795556 1.0372984 0.6872984 2.2120730 3.7226365 2.0698476 0.7190492 0.5522444 0.5664190
 [40] 0.3719857 0.6718746 3.0911603 0.6369714 0.6026667 0.9150175 1.6734794 4.4721587 1.6720286 3.6115635 0.2329190 3.2658333 0.4981397
 [53] 1.5941365 0.3236000 2.3737127 1.2984190 4.2005540 0.9878603 2.6522302 5.0596937 4.5565063 5.8117238 0.8221746 1.5770270 1.5846794
 [66] 0.5568159 0.2652794 1.4443048 0.6429143 0.3205270 0.5664984 0.6867254 0.2321952 0.6210603 0.3590079 1.5557048 2.6830095 0.4459254
 [79] 0.1953032 0.5352143 0.4281889 0.2108286 3.6241540 2.0586698 0.7557619 0.2483413 0.7979381 1.6116143 0.3092873 0.3967794 0.5352794
 [92] 0.9299048 0.1981476 2.9568127 0.3677889 0.5892556 0.6459762 0.2117698 1.5209048 0.3030048 1.6744286 0.6142349 0.4852873 0.8447746
[105] 0.5747683 0.3658254 1.2519302 0.3377603 0.3008873 0.1876952 0.2515556 1.1697841 0.3497365 0.9088175 0.3924857 0.2943095 0.1708794
[118] 1.5822683 0.5152111 1.0858159 0.5671651 0.6615587 0.3354159 2.1299746 0.6106190 0.9888476 0.8892254 0.9164524 0.6929667 1.4456667
[131] 0.6060127 0.4215254 2.4169095 0.5871302 0.9451873 0.9250810 0.8296714 0.6841365 0.4611365 1.6477333 1.9417762 0.7203492 0.2810000
[144] 2.7864683 0.2661714 2.0093810 0.1677603 5.3504127 1.1783889 0.2881349 2.3514952 1.1643762 0.9692429 0.3899508 0.3680921 0.9589016
[157] 0.3468587 0.7413127 1.6709079 0.3227540 1.8913381 1.3232508 0.4703016 1.3748317 0.6598635 0.6180556 0.3317762 0.3665508 0.7298444
[170] 0.7535270 0.7860143 0.3423032 0.3709492 0.2984476 0.6128333 2.6053095 0.1487048 1.8125159 0.8222413 2.9831143 0.6380921 0.4370270
[183] 1.3112714 0.2357317 0.8825571 0.4087667 2.6156175 0.5817857 1.2547397 0.3479984 0.2956984 0.5381937 0.3939270 0.6511032 0.4964698
[196] 0.8491206 0.1825698 0.5660206 0.3909238 1.0347190 0.5059381 4.7185127 0.4297635 0.4176556 1.8326476 1.2210524 1.2303746 0.5488000
[209] 0.5042905 0.3957889 0.8571905 0.9632317 0.4814429 2.8532746 2.4930476 0.5554476 0.3679222 3.1732444 0.2882810 0.3586206 0.7461238
[222] 0.7976270 0.7254540 0.2315683 0.4176937 0.6595905 0.3289048 0.5841254 0.3484730 1.0155429 0.4640540 0.8166492 0.1565254 0.3022651
[235] 1.1295143 0.7542984 0.5168857 0.3283762 1.0028889 0.4573698 2.3157032 0.3131365 0.3490254 0.4750905 1.4762000 1.8758635 0.7244952
[248] 0.7643302 0.6627079 1.1410429 1.1745984 3.4928556 0.4809222 2.7593159 0.4845365 0.8821000 0.6392175 0.2105175 0.2086063 1.4913254
[261] 2.2705968 0.8162683 0.6955286 0.3052857 0.3668984 1.9384952 0.5206825 0.5464556 0.2600333 0.8517508 0.2890857 1.8440270 1.2767111
[274] 0.8307794 0.3945857 1.8075317 0.4913381 1.2168127 0.5308111 0.6062048 1.2740556 0.6396492 0.6282968 0.1514429 1.7176159 1.3936984
[287] 2.6916000 0.2964873 1.0102841 0.3647667 1.2172048 5.2389651 0.9127683 2.7512238 1.1065429 1.4411968 2.4155159 2.7367794 0.3850905
[300] 0.2136111 1.3187190 3.0567587 0.3422222 1.0564603 0.5502635 0.7306032 0.2173349 1.1874667 0.5321984 0.4803698 0.1440016 0.4960571
[313] 0.3787603 0.9036730 0.4288190 1.3832730 2.1450841 1.7147778 0.6357095 1.1606238 0.7964063 0.2806603 0.3920127 0.3360587 0.5298206
[326] 1.6885206 0.4971619 0.2490143 4.1859238 3.9663206 4.3197889 0.7194889 0.5250222 4.4645730 0.5504413 1.9914190 0.8432778 0.4652476
[339] 0.3546524 0.5739952 1.1844254 1.8761333 1.3280000 0.6913730 1.9969857 1.9748651 4.6296556 0.9602508 1.1975810 0.3671397 0.7651635
[352] 0.7447587 2.5419587 0.3983016 3.9018032 0.7201651 0.1591587 0.2930952 0.2231810 3.7402603 1.6059365 0.7345698 0.3201333 3.0515492
[365] 0.5495159 0.5446095 0.2701222 0.8542984 0.8607524 0.9074095 0.6330587 0.7747159 1.1256794 0.8432968 0.4907873 0.5575524 0.3200746
[378] 0.5339746 0.4176619 0.6208190 0.5251429 0.2057397 0.9148460 0.6064841 0.2676254 0.2809841 0.4502619 0.3961556 0.6598587 0.4457778
[391] 0.8372698 0.5140365 0.5655063 0.5919968 0.2265540 1.0595889 0.8866222 0.2148587 0.2328508 0.6858714 0.6104698 2.1962603 0.4583921
[404] 0.2756841 0.6432730 1.6981968 0.4252413 0.6687032 0.4270000 0.2926492 0.2376365 0.6521889 0.2633968 0.5404397 0.6538683 0.6035095
[417] 0.6020476 0.5719286 0.9339698 1.7078778 0.8146921 1.2982698 2.5715095 0.9704270 1.1782397 0.4986175 0.3447492 0.1510762 0.3307762
[430] 2.7111238 0.7222635 0.7248190 0.4379810 0.8568667 0.4940619 0.3673492 0.3505635 1.4963778 0.7446857 0.3997159 0.5405683 0.3361159
[443] 0.2909714 0.6736095 0.7504651 0.7799778 0.6791079 3.7253270 0.5824492 0.7147190 1.2380333 0.4066508 0.6531016 0.3224349 1.5455556
[456] 0.7585444 0.4131302 2.8834159 0.9851667 1.9242698 1.2320937 3.2220746 0.2892984 0.9230873 0.6789905 0.5137603 0.5246984 3.4382619
[469] 3.1764079 2.4343413 1.6549206 1.1268524 1.3680460 0.5851714 2.6817413 1.4130365 0.3852079 0.9323238 1.0584111 0.5777698 1.8673254
[482] 1.3224349 0.4366968 0.3948460 0.5983841 0.3793698 0.5737524 1.1203397 0.7130000 0.6957921 0.8332111 0.6353841 0.9238587 0.3956127
[495] 0.7677571 0.5673778 0.7497762 0.2350111 1.0017714 0.7649413 1.1728238 0.4063873 0.4722810 0.5698175 0.3613159 0.2096825 1.4787683


[508] 0.2055159 3.8660222 1.3550254 0.3288000 0.2751746 0.8064063 0.5592619 0.5496730 0.3874254 0.4136635 0.6418714 0.9658873 4.3452556
[521] 0.3020175 0.6389556 2.3593127 1.4639317 0.2944413 0.2921476 0.5570825 1.9592571 0.5617937 0.9454921 0.4773921 1.1397714 0.4165095
[534] 0.4832857 0.2895841 1.5978635 0.4127921 3.8021698 2.5695095 4.6969254 4.4345984 0.2847190 0.1833587 2.2730952 2.2681889 1.1909968
[547] 2.1417381 0.2721952 1.5640000 0.2123714 4.6479127 0.8101111 0.2285048 0.9699810 0.7318762 0.2282190 0.3468968 0.4623619 0.5164905
[560] 0.9057937 0.5747524 0.8966524 3.5370429 0.5810540 0.1127349 0.9893270 1.1706778 0.5251206 0.6486032 0.8342016 1.3745032 2.4941714
[573] 1.4783698 1.4433238 0.6406825 1.2452286 0.6172159 0.6310540 0.2189000 0.5004190 2.0350206 0.4062762 0.5915190 1.8038524 0.4229063
[586] 0.6595302 0.2376000 2.0901810 0.6462667 0.3618619 0.3555143 0.5649651 2.1247667 1.4677635 1.0007397 1.6328063 1.2322095 0.3980381
[599] 0.7584937 0.3973302 0.9989159 0.8010270 0.5992937 1.4385746 0.8186286 1.5430302 0.8103492 0.2010063 0.7694063 0.5543286 0.4522444
[612] 0.3203984 0.4324921 1.7013048 1.2127540 0.2155016 2.4690381 0.1862222 2.1393508 0.9361349 0.2242587 0.8074095 0.4491825 0.1694302
[625] 0.7264968 0.2260667 0.4256270 0.4813921 0.5995889 0.7498905 1.0101222 0.2520825 0.7152365 1.4960349 0.1670000 0.5542381 0.5562127
[638] 1.5873778 0.5566619 0.9551841 0.3213444 0.3447270 0.6740000 2.3945302 0.8331016 0.5731825 2.0822952 0.3669889 1.9370317 0.7716048
[651] 1.0312032 0.3635952 0.4787857 0.5199841 0.5399381 0.2902349 1.0159794 0.7367492 0.5652317 0.4890397 0.5631508 0.9204746 0.5064651
[664] 0.4345873 1.0512556 0.4293111 0.7636079 0.5684794 2.4782079 0.6623825 0.7947127 2.0154000 0.6476222 1.1377349 0.4653460 0.3859270
[677] 0.4670381 0.4733111 0.9987492 0.4442365 0.4267746 0.6322841 0.4821222 0.6401540 0.8090937 0.8147079 0.4901571 0.5510429 2.4247429
[690] 0.3290286 0.4762444 0.8196492 0.3425492 0.3215333 0.3530000 0.6732460 0.5586603 0.1919159 1.3156048 0.3359905 0.4914317 1.0868714
[703] 0.8311635 0.7145063 0.4234603 0.7649413 0.6324683 0.4866063 1.2099460 0.5131492 0.4466238 0.8720460 0.2825127 0.5036667 0.4802508
[716] 0.8414667 0.6781349 1.3001190 1.3689349 2.0215762 1.3758333 1.3077556 0.1716095 1.7608079 0.9850810 0.1734603 3.1502635 0.7513810
[729] 0.6688508 0.5172651 0.7929222 0.2627111 0.3886397 0.6763222 5.5446206 1.2990889 0.4889032 0.4656444 1.7887254 0.5270952 0.6502444
[742] 1.1728222 0.7140571 0.5362873 0.5685810 0.6894889 0.3691000 1.1724778 0.3198397 0.8135000 1.4011143 1.3488683 0.4750587 0.7524921
[755] 0.3714635 1.2790286 0.3964143 0.9925048 0.3557984 1.4185556 1.7078143 0.9055254 0.2792254 0.3673381 0.2943127 0.4241603 0.1775905
[768] 0.3768063 0.3340683 1.4587921 0.6242190 0.8216873 0.2831524 1.0964048 1.2405968 1.0178111 0.7530476 0.4680556 0.6912302 0.3603032
[781] 1.6688794 0.4851587 0.3506968 0.6048381 0.2606619 0.3280984 1.1001556 0.9004937 0.6174286 0.4583619 0.8042873 0.3816651 0.9475095
[794] 0.5448921 0.6627063 1.0761159 0.9033032 0.6300476 0.3033698 1.2757556 2.6539794 1.2836873 1.0546048 0.8539825 0.4563635 0.6882460
[807] 1.2928794 0.4236349 0.8810651 0.5452095 2.2089603 0.5445079 0.5517254 0.4827032 0.4115476 0.9290190 0.8027048 0.4877476 0.6178778
[820] 0.7650476 0.9498825 2.0868143 0.6508635 0.5126952 0.7559762 1.2502238 0.2987365 0.4914111 0.4175429 0.7037714 1.2099000 1.0849492
[833] 0.3506778 0.5336714 0.9312778 0.5600603 0.4788635 1.9523889 1.3085778 0.6200714 0.7408571 2.3710016 1.0876857 1.4984079 0.3549127
[846] 0.5000746 0.5637492 0.1775746 0.3848873 0.4133206 1.2433492 0.6881746 0.4084937 1.3422492 2.7987746 0.5249381 1.4113032 0.2908095
[859] 1.6098349 1.2639619 0.4219889 0.5959333 0.6546508 1.0313889 1.0106556 2.7225571 0.4142079 0.4227286 2.1229921 0.6265302 0.4247206
[872] 2.3049143 1.5809175 1.7834032 1.1209127 0.4103254 1.5614587 0.3522429 0.3382683 0.4066905 1.1087397 0.5600746 1.5837556 1.0472571
[885] 0.4434667 3.7468905 0.3532794 0.3090889 2.1701159 0.4805571 0.5349873 1.1554429 0.4513508 1.1262683 0.6549317 0.6005746 0.5131270
[898] 0.6597444 1.8565349 0.7568952 2.9137857 0.7919889 0.8251730 0.3742000 0.8323016 0.8775111 0.5476873 0.7786317 0.2776937 1.3970254
[911] 2.0951968 2.3033127 0.3861381 0.3380111 1.4070794 0.9999635 0.2390095 0.4396603 0.1908222 0.3030524 0.2451937 2.1256635 0.6349095
[924] 1.1813413 1.5875206 0.7654063 0.6617206 0.1685651 1.0452238 1.0110873 0.3313889 0.3813079 0.7735429 0.5370556 0.7941683 1.1474063
[937] 2.4256032 0.3213016 0.4680095 0.2811175 1.3332746 0.4340952 2.0554841 0.4308190 1.3646270 0.1549524 0.3284175 1.4527206 1.7966921
[950] 0.7643556 2.3585333 0.7167476 0.5063651 0.7316349 1.1107079 0.8909460 1.3314048 0.4651381 0.8959095 0.2821270 1.2406683 0.6814746
[963] 1.3762524 0.8273794 0.2491714 1.2662032 0.6452476 0.5511556 0.9377460 0.7719794 2.1132587 0.4960937 0.6301968 1.2264524 1.3440540
[976] 1.2772857 1.3579603 1.2671968 1.2578571 0.6378968 0.4283381 1.3222635 1.8718905 0.4493127 0.4088714 0.5098095 0.6963746 0.8023254
[989] 0.9595825 0.3128508 2.7982651 4.2735206 0.8361032 1.0287175 0.7283778 3.9481540 0.8265429 1.2813873 0.4152444 0.4826048
[ reached 'max' / getOption("max.print") -- omitted 1308 entries ]
```

驗證結果是否相同

```
# AI
all.equal(mean_gene_cross_samples1, mean_gene_cross_samples2)
```

Ouput:

```
[1] TRUE
```

如何驗證兩向量所有內容是否相同

審視多種向量比較方法及其應用場景。

在 R 裡驗證兩向量是否相同，會依「相同」的定義不同而有幾種做法。以下由嚴格到寬鬆說明。

1. `identical()` ——最嚴格

檢查型別、長度、順序、屬性全部完全一致，回傳單一的 `TRUE` / `FALSE`。這是判斷「完全相同」最可靠的方式。

```r
x <- c(1, 2, 3)
y <- c(1, 2, 3)
identical(x, y)   # TRUE
```

要注意型別差異它會抓出來：

```r
identical(c(1, 2, 3), c(1L, 2L, 3L))   # FALSE (double vs integer)
```

2. `all(x == y)` ——逐元素比較

先用 `==` 逐一比對，再用 `all()` 確認是否全部為真。比 `identical()` 寬鬆，不管型別細節。

```r
all(c(1, 2, 3) == c(1L, 2L, 3L))   # TRUE
```

但這個寫法有兩個陷阱要小心：

- **長度不同時**會觸發回收（recycling），可能給出誤導的結果，甚至只是警告而非錯誤。保險起見可以先比長度：`length(x) == length(y) && all(x == y)`。
- **含有 `NA` 時**，`==` 會回傳 `NA`，導致 `all()` 也變成 `NA`。可以加 `all(x == y, na.rm = TRUE)`，但更穩健的還是用 `identical()`。

3. `all.equal()` ——容許浮點誤差

比較數值時，因為浮點運算的精度問題，`==` 有時會失準。`all.equal()` 允許微小的容差：

```r
all(0.1 + 0.2 == 0.3)        # FALSE！浮點誤差
isTRUE(all.equal(0.1 + 0.2, 0.3))   # TRUE
```

注意 `all.equal()` 在不相等時回傳的是「差異的描述文字」而非 `FALSE`，所以通常要包一層 `isTRUE()` 來取得乾淨的邏輯值。

Get sd value for each gene cross samples

```
# 求 sd
sd_gene_cross_samples <- apply(bb, 1, sd)
sd_gene_cross_samples
```

Ouput:

```
  [1] 0.89450660 0.36384650 0.64768521 0.59063392 0.17078522 1.73313507 2.29676297 0.57714540 0.61919966 0.68873732 2.11331157 0.36123722
 [13] 1.75226687 0.80921771 0.57810004 0.45068305 0.42105331 0.99973280 0.89446868 0.64282018 0.72289529 1.96578456 0.26724024 1.87366765
 [25] 1.54412319 1.58982814 0.73400744 0.73236215 0.24418548 0.76446123 0.27688691 0.55746782 0.88040157 0.86094315 1.37375784 1.07871957
 [37] 0.43145407 0.35214699 0.23721084 0.19596735 0.70055764 1.14237882 0.58407683 0.38559346 0.56886954 0.82176548 1.49884810 1.06178497
 [49] 1.78295187 0.18666423 1.44506318 0.25413569 0.69003412 0.28770547 1.16823223 0.64971758 1.19211363 1.06431709 0.89634741 2.15836904
 [61] 1.64198717 2.94158419 0.34631338 1.06734895 0.78725963 0.58166063 0.21381817 1.71290689 0.22538342 0.31579080 0.33186143 0.43350623
 [73] 0.22805344 0.60539041 0.33473733 0.86205502 1.68657181 0.22531985 0.10044619 0.24437080 0.19586322 0.07439796 1.41602045 1.13542581
 [85] 0.88184578 0.16080598 0.71801756 1.27282750 0.21334216 0.33425097 1.02733665 0.52302379 0.16041734 1.04303878 0.21754677 0.24878733
 [97] 0.24948388 0.14090053 0.86595588 0.23322820 1.25485462 0.36285418 0.22077619 0.41732726 0.20315274 0.21567213 1.17333010 0.34941396
[109] 0.15322944 0.06922589 0.13401291 1.03953847 0.12927031 0.43011423 0.64098721 0.18713109 0.11033816 0.82456558 0.69027282 0.40114900
[121] 0.25652759 0.47457602 0.30327206 1.26185057 0.29570234 0.68878906 0.89389467 0.42138371 1.19137355 1.09064437 0.32553016 1.05701188
[133] 0.97438367 0.92010822 0.63710952 0.64811349 0.61102788 0.35256927 0.44208162 0.82327106 1.80375545 0.56744140 0.20560509 1.32478146
[145] 0.23819976 2.18553914 0.09231798 1.98293698 0.66826500 0.19091036 2.02441944 0.65199824 0.79783170 0.42868781 0.19518101 1.33485595
[157] 0.51244873 0.33764144 0.76666250 0.21337665 0.73005978 0.84690835 0.14617160 0.99668049 0.60119002 0.53205846 0.22379809 0.13639605
[169] 0.98825051 0.47202026 0.38207599 0.20167681 0.15009537 0.32517386 0.34662904 0.98207443 0.10530056 0.75367043 0.44506650 1.29191149
[181] 0.38837974 0.28544894 0.86915800 0.71627830 1.03713441 0.27236458 3.91623547 0.48434617 1.03759996 0.34759613 0.16542389 0.64622682
[193] 0.39370508 0.45048568 0.39770046 0.48652890 0.09860386 1.45377070 0.20877360 1.07829075 0.44880703 1.51106376 0.28797168 0.40720225
[205] 0.95833932 0.62758318 0.68509825 0.25251461 0.30615374 0.20362659 0.37824746 0.61575212 0.27717058 1.00503825 0.93584691 0.39820303
[217] 0.46912633 1.27932473 0.22591672 1.13303221 0.32563900 0.42767445 0.34961187 0.21012444 0.26772260 0.30775808 0.15115679 0.42479306
[229] 0.20868188 0.93819449 0.63982837 0.97023952 0.07526037 0.28538790 1.46910685 0.83154689 0.45924072 0.20696114 0.43545371 0.19939099
[241] 1.04798923 0.16323045 0.15203648 0.38768982 0.57525795 2.12252824 0.69506007 0.42218417 0.38543106 0.68102620 1.26844337 1.25059868
[253] 0.30813016 1.52850158 0.59586277 0.71767184 0.60294904 0.22452451 0.09670972 0.72199317 1.47905395 0.30358631 0.39524228 0.20822964
[265] 0.22455655 1.34479642 0.55338317 0.27929372 0.12771868 0.39255790 0.17646918 1.40547744 0.68306273 0.55814519 0.20626197 1.78687732
[277] 0.26369249 0.68676269 0.22621581 0.42249639 0.93203254 0.41839024 0.28564804 0.06589355 0.76644648 1.05401164 1.04823323 0.14059661
[289] 0.95414333 0.24140695 0.57003301 2.08110661 0.40217276 1.58666452 0.86117375 0.71398504 0.97828766 1.14029795 0.28689374 0.17568324
[301] 0.68596541 1.45848297 0.31297999 0.39643102 0.27407407 0.37105971 0.09434923 0.82022577 0.32185689 0.35988239 0.09559869 0.40795193
[313] 0.16912404 0.61281316 0.25151143 0.75752835 0.98593103 0.50993837 0.31064679 0.66285473 0.54136864 0.20873102 0.43296143 0.19075394
[325] 0.44093642 1.61943947 0.20830196 0.17747119 2.08556144 1.58498728 1.63054883 0.56865276 0.41351387 1.92092537 0.60421185 1.26915359
[337] 0.42953504 0.46408830 0.14058002 0.29695357 0.71081696 0.75353490 0.60430954 0.43985652 1.08409941 0.91046611 1.53512986 0.87210064
[349] 1.56382171 0.36576119 0.46784250 0.79832880 1.02232395 0.36165115 1.24661349 0.27000216 0.08989501 0.16466148 0.16248218 1.80299566
[361] 0.58474038 0.62394445 0.15348710 2.10618122 0.42136022 0.24128597 0.11691393 1.01080395 0.56760539 0.68467723 0.76552212 0.43451018
[373] 1.00482237 0.46749158 0.39760202 0.39968549 0.19129442 0.32649830 0.21649722 0.62156300 0.63304737 0.11979722 0.37920113 0.40497400
[385] 0.16542793 0.13612399 0.14838870 0.17702655 0.31309085 0.25563947 0.46300750 0.29956236 0.28527269 0.23847343 0.38356639 0.58787401
[397] 0.35286206 0.07243994 0.13673577 0.41190084 0.28365818 1.08977947 0.28334553 0.22121580 0.40974336 0.71652272 0.48381290 0.56099367
[409] 0.33290144 0.11896839 0.13204255 0.79781523 0.12893745 0.23733856 0.52077206 0.56676555 0.55941579 0.37010394 0.46942677 0.76631699
[421] 0.35250122 0.71387405 1.21275429 0.27865911 0.54334013 0.23072387 0.14958071 0.07578075 0.12943397 2.36165823 0.37242929 0.29066033
[433] 0.49591531 0.81761854 0.26806445 0.18350797 0.29009943 0.91368667 0.31400946 0.17108525 0.41571643 0.23778143 0.16968102 0.37969119
[445] 0.43570340 0.32629702 0.34006343 1.39495871 0.39382524 0.34358398 0.49848297 0.33312604 0.25820672 0.30386676 0.74406179 0.42917900
[457] 0.18705958 0.94692927 0.55576494 0.82944453 0.52879500 0.99251803 0.09616998 0.40755119 0.43430270 0.28150106 0.30283523 1.33068035
[469] 2.04973094 1.17105244 1.03405410 0.81300652 0.67192633 0.40105249 1.30359326 0.65292124 0.24004472 0.43252214 0.39277394 0.36771812
[481] 0.57852173 0.41562177 0.29667296 0.18172724 0.48296276 0.18180944 0.20910142 0.81027436 0.34365994 0.38143594 0.44318348 0.26219820
[493] 0.54140922 0.35106274 0.53553680 0.25735489 0.29607579 0.12873492 0.54168134 0.39303263 0.84372034 0.24634474 0.43806390 0.38066441

[505] 0.17028082 0.11414320 0.61582880 0.12803561 5.57821100 0.80447624 0.20201838 0.16439086 0.43874031 0.31016725 0.35021805 0.20460547
[517] 0.22627803 0.75242527 0.32315256 2.04442019 0.23668600 0.82042823 1.95591644 1.15837355 0.23857481 0.22905889 0.29887742 0.79311755
[529] 0.51761715 0.47998632 0.58699945 0.66086475 0.39874533 0.27951461 0.16594704 1.34010332 0.18358670 1.11794208 1.09098324 1.87433971
[541] 1.33936301 0.46174556 0.12697814 1.58883172 1.97217941 0.61966282 1.13134294 0.16208429 0.68546877 0.08958473 2.39012321 0.43272007
[553] 0.16194266 1.81415719 0.31485193 0.12635811 0.24980409 0.32025757 0.28877976 0.86826309 0.33352114 0.38942594 1.40372098 0.22790674
[565] 0.08196502 1.58267067 0.98442690 0.29456823 0.45409867 0.36788793 0.66236865 1.58335284 1.18272164 0.90720289 0.45635665 0.87305874
[577] 1.08301737 0.31198475 0.12077394 0.31511581 0.98568571 0.20057874 0.31888142 0.89093276 0.26706231 0.37846581 0.15214978 0.78081111
[589] 0.55034032 0.25904993 0.20183770 0.31286359 1.16214456 0.94319734 0.39616491 0.97506073 0.66948814 0.30718367 0.35353105 0.21872347
[601] 0.57175146 0.41915493 0.68437952 1.13244496 0.37727704 0.99381484 0.61074864 0.15612748 0.27508288 0.32990328 0.28368683 0.24955745
[613] 0.30565184 0.75722964 0.60925428 0.10837787 0.90564382 0.08509329 0.78393747 0.41102439 0.17977858 0.51425369 0.33356803 0.17121902
[625] 0.64513741 0.11123410 0.32772364 0.33569737 0.36358931 0.54974338 0.82539787 0.13505848 0.64005803 0.68440780 0.12244251 0.34902034
[637] 0.27261414 0.67595644 0.33194896 0.47414109 0.15236534 0.34747023 0.36980676 1.33177061 0.41308475 0.32427365 0.98220617 0.14779664
[649] 0.90852368 0.43702718 0.47986969 0.17660262 0.33535341 0.39124005 0.47850609 0.16499203 0.88911253 0.30072912 0.44159038 0.24980515
[661] 0.25176009 0.53881716 0.35455859 0.15500259 1.12163115 0.35529819 0.61785007 0.42443000 1.17521963 0.34119344 0.82174005 2.77851423
[673] 0.35280050 0.47227641 0.38003673 0.20243068 0.23615033 0.22269342 0.47128998 0.19349393 0.30992897 0.81200948 0.25617641 0.23613971
[685] 0.52830437 0.43628484 0.46042055 0.44611552 1.56766600 0.19829435 0.22760838 0.42028200 0.23726723 0.22949255 0.39783030 0.55837166
[697] 0.47608890 0.08986528 0.94710938 0.19015032 0.21614905 0.42758213 0.41066474 0.34838531 0.21704761 0.42283109 0.55501315 0.21751385
[709] 0.92377259 0.28149969 0.34657001 0.35740748 0.14658391 1.40429165 0.35047634 0.34071346 0.47465970 0.49497036 0.89322014 0.85570535
[721] 1.00837093 0.57050056 0.06856860 0.78896108 0.48601331 0.12954546 1.32834626 0.36466268 0.61370978 0.63325877 0.47764834 0.13192192
[733] 0.24069076 0.44225859 2.60699797 0.73715777 0.33468598 0.24982144 0.85681902 0.28819586 0.79256022 1.23266934 0.47733994 0.30340836
[745] 0.34831241 0.37735855 0.15557222 0.75076559 0.15621282 0.36576037 0.74676036 0.84572453 0.45158504 0.47927233 0.28534289 0.69005525
[757] 0.16265232 0.95564326 0.20168846 1.00372674 1.49208179 0.78195686 0.13938746 0.26998402 0.14980777 0.37396168 0.09944945 0.17205680
[769] 0.20931866 0.56862994 0.26260586 0.33887850 0.19297868 0.66686355 0.86055753 0.77721570 0.50722465 0.23463696 0.63223233 0.19982656
[781] 1.17441387 0.30940283 0.46805631 0.26521594 0.10863321 0.14752793 0.58791713 0.62070673 0.29556050 0.27050403 0.41916039 0.20640094
[793] 0.47666547 0.30559311 0.31118520 0.65102477 0.52700293 0.35191793 0.25683405 1.13581389 1.20904356 0.68565127 0.49437436 0.37921171
[805] 0.26664248 0.37872002 0.55595944 0.38874239 0.43889372 0.32568777 1.16044825 0.27293751 0.24786716 0.25994949 0.33656246 0.52310623
[817] 0.60344048 0.28333998 0.65924179 0.51593996 0.33920329 1.00212814 0.38632147 0.63736642 0.39046149 1.39652218 0.15197134 0.17387949
[829] 0.47988534 0.37588602 1.90972151 0.36014301 0.32771815 0.20421980 0.49803902 0.78895543 0.35767403 0.81482872 1.11522834 0.28192130
[841] 0.31462277 1.55551914 0.41766689 0.93413194 0.22957322 0.69609862 0.45601064 0.16524616 0.27522802 0.32464178 1.38601167 0.35843872
[853] 0.24990894 0.51342980 1.18361677 0.34581757 0.65996016 0.33080998 0.84435078 0.51857442 0.25753251 0.36461919 0.26963946 0.57455313
[865] 0.50886572 1.00710317 0.37780724 0.19943694 0.94676661 0.35955230 0.20317695 1.18695731 0.68603512 1.19787998 0.58864259 0.18541754
[877] 0.77284656 0.22488097 0.16857208 0.45024194 0.47789624 0.23995559 0.89974518 0.41146404 0.22116324 1.41375055 0.23502925 0.36264760
[889] 1.02488065 0.25245458 0.30757972 0.46156727 0.19964119 0.53876307 0.28981903 0.35729009 0.29170653 0.47484593 0.71883087 0.52882935
[901] 1.19565728 0.37784672 0.39875801 0.82679248 0.99478003 0.42618879 0.20370572 0.50101278 0.33458592 1.18315376 1.08509438 1.16820941
[913] 0.18406378 0.13354359 0.71143788 0.51212341 0.11342121 0.23493651 0.14031269 0.20019234 0.15822426 0.87905919 0.30749942 0.84534168
[925] 0.89733242 0.53622661 0.24668914 0.09618566 0.98400094 0.60851422 0.22958982 0.38018437 0.64017556 0.23763017 0.42471756 0.64219307
[937] 1.37188314 0.16291218 0.25393548 0.18401608 0.71622969 0.28266080 0.68905395 0.25373290 0.69248850 0.20085426 0.17439956 0.55476848
[949] 0.68700663 0.40140297 1.39609252 0.63941264 0.27211782 0.46531964 0.48890522 0.36569908 0.62957350 0.16249412 0.36188196 0.14000732
[961] 0.51174390 0.34354398 0.81295160 0.37056400 0.24024224 0.51537506 0.31677085 0.22930581 0.46062073 0.82375307 0.94165919 0.28426809
[973] 1.24177510 0.85137914 0.81019180 0.85074369 0.55711375 0.69409130 0.90821784 0.67488672 0.27631160 0.50511178 0.98002643 0.43146771
[985] 0.14757107 0.25257319 0.26371707 0.56327695 0.68943587 0.28332573 1.58121432 1.77009029 0.30907717 0.43726395 0.58534132 1.83224567
[997] 0.43113452 0.73069240 0.26159713 0.26664387
[ reached 'max' / getOption("max.print") -- omitted 1308 entries ]
```

題目二：Get mean & sd value for each sample cross genes

Get mean value for each sample cross genes

```
# 題目二：Get mean & sd value for each sample cross genes

# 求 means
# 方法一
mean_sample_cross_genes1 <- apply(bb, 2, mean)
# 方法二
mean_sample_cross_genes2 <- colMeans(bb)
```

Ouput:

```
[1] 0.8999231 0.9107066 0.9121858 0.9216109 0.9393284
0.9246767 0.9181976 0.9211571 0.9080536 0.9379128 0.9000853
0.9039657 0.9176309
[14] 0.8991331 0.8911080 0.8915395 0.9212825 0.9070417
0.9039065 0.8974366 0.8833335 0.8983932 0.8853999 0.8999410
0.8912325 0.8921722
[27] 0.8891327 0.8815866 0.8788212 0.8820618 0.8865663
0.8988042 0.9210532 0.9083504 0.9061526 0.8964495 0.9260049
0.9098169 0.9065384
[40] 0.9023300 0.8988161 0.8996560 0.8955150 0.9109578
0.9330225 0.9104899 0.9240139 0.9156453 0.9120389 0.9096284
0.9137515 0.9013536
[53] 0.9162928 0.9240076 0.9189305 0.9310080 0.9385528
0.9221431 0.9259349 0.9295506 0.9279080 0.9167595 0.8906594
```

驗證結果是否相同

```
all.equal(mean_sample_cross_genes1, mean_sample_cross_genes2)
```

Ouput:

```
[1] TRUE
```

Get sd value for each sample cross genes

```
# 求 sd
sd_sample_cross_genes <- apply(bb, 2, sd)
sd_sample_cross_genes
```

Ouput:

```
   EWS.T1    EWS.T2    EWS.T3    EWS.T4    EWS.T6    EWS.T7    EWS.T9   EWS.T11   EWS.T12   EWS.T13   EWS.T14   EWS.T15   EWS.T19
0.9023934 0.8952745 0.9059050 1.1925464 0.8663332 0.9343484 0.8483575 1.0391300 1.1529887 1.3418554 1.1775772 1.2658881 1.2677206
   EWS.C8    EWS.C3    EWS.C2    EWS.C4    EWS.C6    EWS.C9    EWS.C7    EWS.C1   EWS.C11   EWS.C10     BL.C5     BL.C6     BL.C7
1.1612246 0.9871340 0.9954100 0.7455796 0.8318380 0.9232795 1.0691206 1.0979321 0.9725914 1.1268339 0.9649714 0.9442416 1.0255234
    BL.C8     BL.C1     BL.C2     BL.C3     BL.C4     NB.C1     NB.C2     NB.C3     NB.C6    NB.C12     NB.C7     NB.C4     NB.C5
1.0811283 1.0854579 1.1259233 0.9222812 0.9586906 0.9186860 0.8783569 1.1563298 1.0061901 1.0522149 0.9551821 0.8653178 0.9557501
   NB.C10    NB.C11     NB.C9     NB.C8    RMS.C4    RMS.C3    RMS.C9    RMS.C2    RMS.C5    RMS.C6    RMS.C7    RMS.C8   RMS.C10
0.7951259 0.9779185 0.9523493 1.1118753 1.2229604 0.8485316 1.4133762 1.2185764 0.9787163 0.8492464 1.0370480 0.8156487 1.1493356
  RMS.C11    RMS.T1    RMS.T4    RMS.T2    RMS.T6    RMS.T7    RMS.T8    RMS.T5    RMS.T3   RMS.T10   RMS.T11
0.7998934 0.9975282 0.9611903 0.8886484 1.1732700 1.7319878 1.0980215 1.0859599 1.0401830 1.1738220 1.6555917
```

Select top 30% genes with larger sd values

```
# AI
n_top30_cols <- ceiling(nrow(bb) * 0.3)
top30_genes_sd <- order(sd_gene_cross_samples, decreasing =
TRUE)[1:n_top30_cols]
top30_genes_sd
```

Ouput:

```
  [1]  509 1834  187 1781 1750 1774   62 1572 1547  672 1897  735 1574 1975  551  430    7 1517  146   60  246   11  364  329  292  469  520  151  148  545
 [31]   22  523  334 1544  831 1372  540   24 1955 1645  996  554  141  360  276   49  992 1708   13 1065    6 1954   68   77 1345 1389   61 1915  331  326
 [61]   26  544  294  330  572  566  991 1826  689 2099  349 1786  842 1653 1600   25 1093 1907  347 1319  254 2223  202   47 1573  761 1079  261  235 2046
 [91]  302 1625  198 1448   51 1977 1070 1546 1621   83  886 2147  272  714  563 1044  826  951  448 2115  851 1771   35  937 1083 1540 1831 1851 1614  266
[121] 1594  536  541 1371  156  644  468  727  144 1072 1765 1927 1074 1478  475 1412  180 1932 1630 1947 1894  218 1299 1739   88  336  251 1113  124 1820
[151] 2290  101  252 1175 1222  355 1890  973  742 1150 1484  423  801 1626  874  901   57  129  872  855  910  573 1607 1243 1227  669 1682  781  107  470
[181]   55  912  593  811  524 1855 1084 1582 2135 1280   42 1615  298 1109 1871  800   84  220  604  547 1295  665 1648 1715 1105  538 1767  839 1256 1076
[211] 2230 2213 1339 1055  539  130  402 1031 1327 1277 1023  911  345  577 2077 1128   36 1021  200 1489 1841 1937   64   58 1353   48 2240 1965  132 1916
[241]  286 1602 1632  287 1601  241 1721 1810 1167 1483   94 1618 2292 1223  112  189  185  471   91 1529 1543 1185  889 1066  353 2273 1772 1090  368 1358
[271] 1497  721  866 1524  214  373  760  822   18 1114 1857  164 1603  905  606 1535  462  169  317  581  567  929  647  176 2277  983  297 1418  596  133
[301] 1169  232 2022 1135 1151 2117  205 1057 1397  758  289  699  458  869 1300 1426 2050  594 1075 1666  971 1961  230 1945  215 1843  844  281 1053 1872
[331] 1494  709 1980 1738  134 1382  438 1814  346  649 1048  979 1183  574  617 1166  883  925 1769 1349 2029   59    1   19  127  719 1900  584 1940  657
[361] 1417 2228 1956 2020   85 1759   33 1307  922 1542 1780 1219 2234  576  348  183  560 1616   99 1803 1654   76  295   34  775 2263  739 1162  720 1885
[391] 2199 1311 1960  974  976 1271  162  752  924  859 1736  501 1764 1067 1973 1748 1690  236 2218  460  904 1116  631 1683  118  970 1795  140 1024 1726
[421]   46  671  522  308  434  838  472  963  682  488  975   14 1730 1812 1310 1229 1752  510 1022 1799 1783  352 1588  153  412  528  741 2220 1972 1623
[451]  724  836 1620 1029   65 1525  619 1596 1568  762 2239 1177  588 1884 1651 1213  776 1858  877 1452 1612 1318 1604 1913 1158  159 1792  285  420 1141
[481] 1883  371   30 1309 1362  316  614 1718 2256  178  342  518  748 1032 1197  751 1901  455 1863 1986 1874 1391 1650  736 1052   27 1837   28  998  161
[511] 1692 1071 1608 2157 1272 1304 1207   21 1282  260 1414  899 1017 2260   87  256 1519  406  184  941 1813 1611 1821  296  422 2148  915 1548  341 1555
[541] 1541 1298 1144 1208 1981   41 1297 1724 1785 1822  846  247  978 1580 1237  945 1139  119 1447  756   53  989  943  126   10 1042  949  278 2171  873
[571]  301  802  549  207  370  634  603 1468  273  250 1725  638  980 1847 1886  473 1533 2303 1924  597 1069  149 1062  774 2221  320  571 1473  532 1094
[601] 1203 2270 1305  857  819 1374 1154 1892 2007 1678  476 1836  152 1402 1844 1441  796   56 1950  136    3  192  625 1015 1260 2139   20 2249  936  115
[631]  933  633  231 2048  952 1922 1025 2113  824  135 1235 2093 2102 1020  730  381 1028  779 2231 1340 1082  957 1281  206 1959 1258 1122  362 1528 1714
[661] 1224  380 1099  788  546 1125    9 1323 1689  667 1200 1416 1550  507  212 1966 1003  729  314 1379 1342  137  607  615  930 2182 1039   74 1644  343
[691]  335 1078 1978
```

ceiling是什麼函式

下午2:01

`ceiling()` 是 R 的**無條件進位**函式，把數值往上取到最接近的整數。

```r
ceiling(4.1)    # 5
ceiling(4.9)    # 5
ceiling(4.0)    # 4
ceiling(-4.1)   # -4（往上取，所以是 -4 而非 -5）
```

負數要特別留意：「往上」是指往數線的正方向，所以 `-4.1` 進位後是 `-4`。

一組相關的取整函式對照：

- `ceiling(x)`：往上進位 → `ceiling(4.1)` 得 `5`
- `floor(x)`：往下捨去 → `floor(4.9)` 得 `4`
- `round(x)`：四捨五入（實際上是「四捨六入五成雙」的銀行家捨入）→ `round(2.5)` 得 `2`
- `trunc(x)`：直接去掉小數部分 → `trunc(-4.9)` 得 `-4`

一個常見用途是計算「需要多少組」，例如 23 個東西每組 5 個要幾組：

```r
ceiling(23 / 5)   # 5
```